End-to-End Pipeline Report

Generated: 2026-05-03 18:51:07
Pipeline: Client → network → NFQUEUE (firewall) → broker → `gateway.py` (PQC re-encrypt) → cloud-ready
Source: `mqtt_firewall.jsonl` + `reencrypt_metrics.csv`

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1. Traffic Summary

Metric	Value
Observation window	09:37:26 → 09:48:11
Packets intercepted by firewall	4,156
Flow verdicts issued	35
Attack flows detected	34 (97.1%)
Benign flows	1 (2.9%)
Benign messages re-encrypted	144
PQC algorithm (gateway)	`ML-KEM-768` + `ML-DSA-65`
PQC mode	Real liboqs

2. Firewall Performance — Per Attack Type

> **Det. Rate** = mean model confidence on attack-classified flows. **Infer** = `model.predict()` only. **Classify** = feature extraction + inference + decision. **Inspect** = packet arrival → verdict (inline only).

Attack Type	N	Det. Rate	Infer μ (ms)	Infer σ (ms)	Classify μ (ms)	Classify σ (ms)	Inspect μ (ms)
MQTT-DDoS-Connect_Flood	2	84.7%	0.402	0.074	1.039	0.079	1.296
MQTT-DDoS-Publish_Flood	16	100.0%	0.574	0.258	1.352	0.653	1.635
MQTT-DoS-Publish_Flood	9	100.0%	1.354	1.627	3.133	3.510	3.472
TCP_IP-DDoS-UDP1	6	82.5%	0.399	0.114	0.883	0.296	—
TCP_IP-DoS-TCP	1	100.0%	2.900	—	6.852	—	—
benign	1	—	0.378	—	1.001	—	1.158
Overall	35		0.795		1.859		

3. Gateway PQC Re-Encryption Timing

> KEM: `ML-KEM-768` | Signature: `ML-DSA-65` | PQC mode: Real liboqs

Stage	N	Mean (ms)	Std (ms)	P95 (ms)	Min (ms)	Max (ms)
	0	0.000	0.000	0.000	0.000	0.000

Stage	N	Mean (ms)	Std (ms)	P95 (ms)	Min (ms)	Max (ms)
WAN transit (optional, client clock)						
KEM encapsulate	144	0.047	0.019	0.065	0.031	0.215
AES-256-GCM encrypt	144	0.030	0.143	0.024	0.005	1.609
Signature sign	144	0.113	0.052	0.221	0.049	0.286
Total re-encrypt (KEM+AES+Sign)	144	0.189	0.176	0.297	0.086	1.979
On-device gateway (on_message→done)	144	0.488	0.981	0.613	0.185	11.477

4. Path Latency (means)

> **Non-overlapping segment means** along the benign path. WAN uses the client's `t_publish_ns` vs gateway `time.time_ns()` (clock skew possible). Gateway handler = `t_recv_to_done_ns` (includes key setup, JSON routing, and timed KEM/AES/sign). Firewall verdict timings are **flow-level** (JSONL), not joined per MQTT message.

Segment	Mean (ms)	Note
Gateway handler (on_message → re-enc done)	0.488	WAN column zero — set `GW_REENCRYPT_LOG_WAN_TRANSIT=1` to log WAN

Inside gateway handler (informational)

> Sub-stages are **subset** of the handler row above (KEM+AES+sign timings ⊆ re-encrypt ⊆ full handler). Do **not** add these to WAN to form a second "E2E" — use the **Sum** row in the table above.

Component	Mean (ms)
KEM encapsulate	0.047
AES-256-GCM encrypt	0.030
Signature sign	0.113
Timed re-encrypt (KEM+AES+Sign)	0.189
Approx. rest (keys, parse, …) ≈ GW − re-encrypt	0.299

Firewall LightGBM mean inference (verdict events): **0.795 ms** — not additive with the path table; packets are accepted before classification, but slow inference can delay **subsequent** NFQUEUE packets.